



BeanHealth

CLR Screening Tool

Phone-based AI triage for strabismus — no ophthalmologist required

Strabismus (squint) affects **1–4% of children** worldwide. Early detection is critical — but access to ophthalmologists is scarce, expensive, and unevenly distributed.

1–4%

Children affected globally

Strabismus is one of the most common paediatric eye conditions — millions undetected each year.

6 mo

Average diagnosis delay

Missed early detection leads to amblyopia (lazy eye) — a window that closes by age 7.

70%

World lacks eye care access

Rural and low-income communities have near-zero access to paediatric ophthalmology.

Current screening tools require clinic visits, specialist equipment, and trained clinicians — making population-level screening impossible.

The Corneal Light Reflex test — a 100-year-old ophthalmology standard — reimaged for any smartphone.

AI

MediaPipe Iris Tracking

Google's MediaPipe FaceMesh detects iris landmarks at 30fps in-browser. No specialist camera — any phone torch creates the corneal light reflex.

CV

10-Frame Ensemble Analysis

FastAPI backend processes 10 frames, applies IQR outlier rejection, and aggregates results — eliminating motion blur and single-frame noise.

Δ

Asymmetry-Based Classification

Uses inter-ocular CLR asymmetry (not absolute displacement), cancelling the natural kappa angle. Eliminates false positives on normal eyes.

°

Hirschberg Conversion

1mm CLR displacement $\approx 7^\circ$ ocular deviation. Normalised by iris radius for scale invariance — works at any camera distance.

ICD

Clinical-Grade Output

ICD-10 coded results, four-tier urgency triage (URGENT / ROUTINE / MONITOR / NORMAL), and a referral recommendation with timeframe.

∞

Zero Hardware Cost

Works on any modern smartphone with a front camera and torch. No app download — runs in a mobile browser. Deployable in minutes.

Clinical Standard: Hirschberg Test · ICD-10 Coded · INCONCLUSIVE rather than wrong — never returns a false triage result



Feature	Cover Test	Photoscreener	BeanHealth CLR
Cost Per screening	—	\$5,000–50k device	Free — any phone
Specialist required	✗ Trained clinician	~ Technician	✓ Parent / nurse
Objective measurement	✗	✓	✓
ICD-10 coded output	✗	~	✓
Remote / home use	✗	✗	✓
Kappa-angle corrected	✗	✗	✓ Asymmetry-based

HOME

Home Screening by Parents

Parents use the app on their phone to screen children between GP visits. Simple flow: enter name, enable torch, hold up to eyes, get result in seconds. No medical knowledge required.

SCH

School Vision Programmes

School nurses run batch screenings during routine health checks. Export results to a triage list — children flagged ROUTINE or URGENT get fast-tracked to an ophthalmologist.

GP

GP Pre-Referral Assessment

GPs use CLR screening before deciding whether to refer — reducing unnecessary specialist appointments by up to 60%. ICD-10 codes slot directly into EMR systems.

TELE

Telehealth & Remote Clinics

Ophthalmologists in remote consultation can request a CLR screen to be run during the call. Annotated result image sent back in real time — clinical decision support without physical presence.

\$9.2B

Global Vision Screening Market

CAGR 7.4% · Projected 2030

200M+

Children Unscreened Annually

Primary target — zero current digital solution at scale

1.4B

Smartphone Users — Target Region

SE Asia + South Asia · High paediatric population density

Revenue Model

- **B2B SaaS — Health Systems**
Per-screen or per-seat licensing for hospitals, school health authorities, and telehealth platforms
- **API Access — EHR Integration**
Embedded CLR analysis API for EMR vendors — fee per API call with volume tiers
- **Consumer Subscription**
Freemium app with pro tier — detailed reports, history, paediatrician sharing
- **Data & Research Partnerships**
Anonymised, consented CLR dataset — valuable to ophthalmology AI research

PHASE 1 Now → 3 months

- Clinical validation — 50 paediatric subjects
- IRB ethics approval for pilot study
- Sensitivity / specificity benchmarking
- Glasses + dark iris robustness
- Mobile PWA deployment

PHASE 2 3 → 12 months

- Partner with 3 school health authorities
- EHR integration API (Epic / Cerner)
- Multi-language support (Tamil, Hindi, Malay)
- Telehealth platform partnerships
- Seed funding round

PHASE 3 12 → 36 months

- CE Mark / FDA 510(k) pathway
- Expand to amblyopia + cataract screening
- Government health programme contracts
- Series A fundraising
- Regional HQ — Singapore / KL

Let's make early strabismus detection universal.

BeanHealth · EyeQ Innovate Hackathon 2.0 · beanhealth.app

BeanHealth